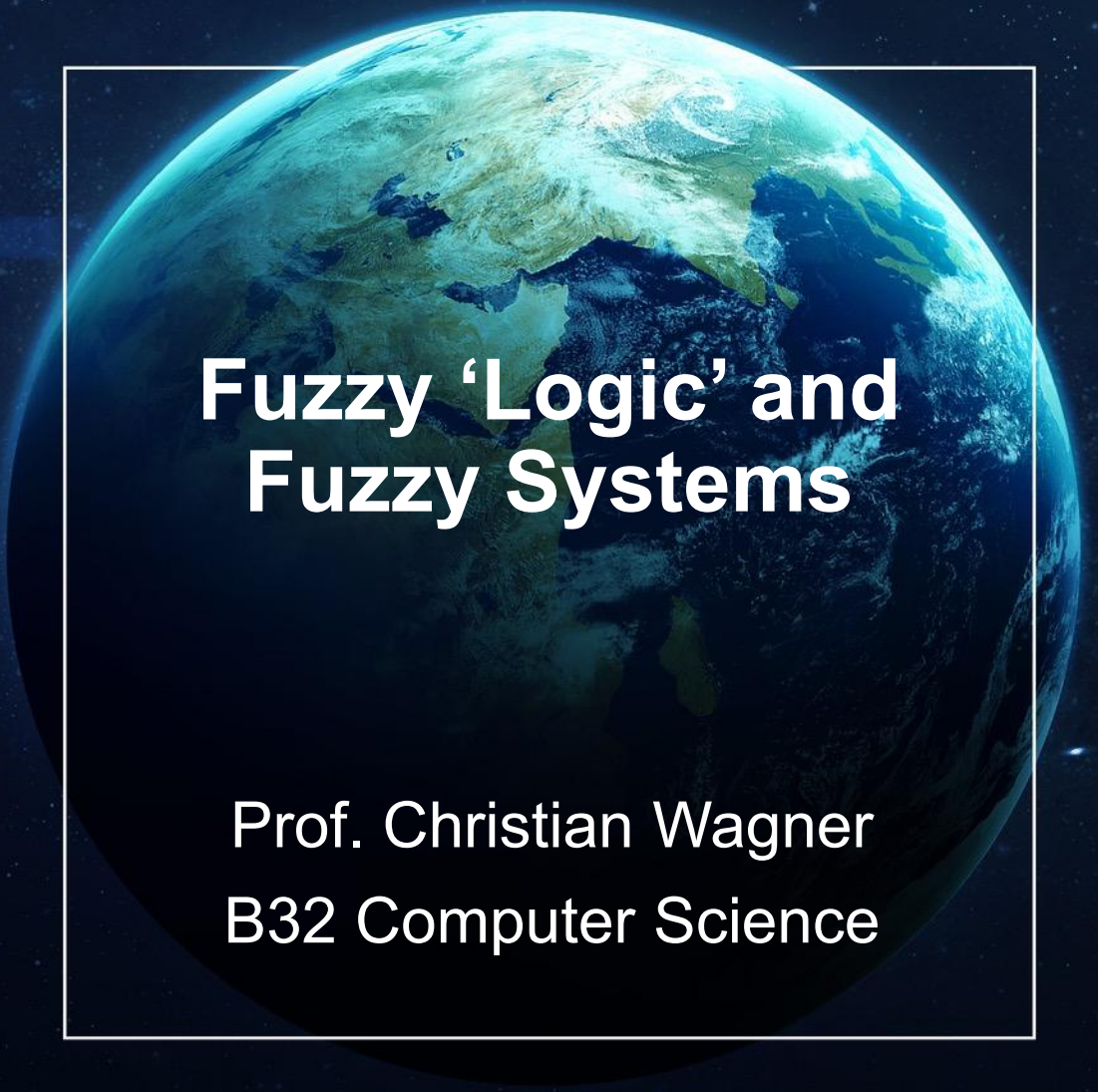




University of
Nottingham

UK | CHINA | MALAYSIA

A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere with North and South America. The image is framed by a thin white border.

Fuzzy 'Logic' and Fuzzy Systems

Prof. Christian Wagner
B32 Computer Science

Overview (Guest) Lecture
COMP2011

Full course: COMP4033



Christian Wagner

Professor of Computer Science

Originally from Echternach, Luxembourg

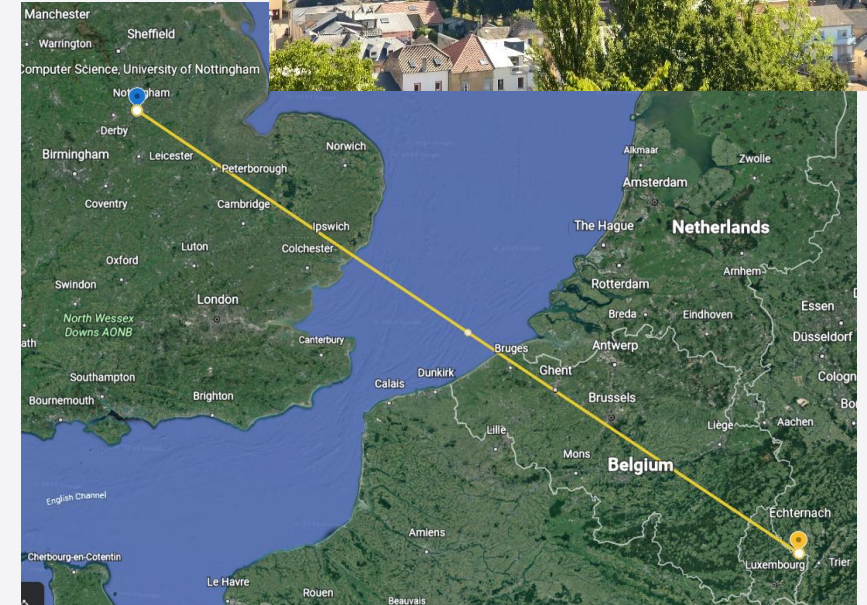
- BSc CS, MSc Robotics & Embedded Systems, PhD Computer Science, Post Doc in Intelligent Homes

@Work:

- Lead LUCID – the Lab for Uncertainty in Data and Decision Making at the School of Computer Science, UoND
- IEEE Computational Intelligence Society, Nottingham Horizon interdisciplinary research centre and UK Trustworthy Autonomous Systems Hub (TAS)
- Research on end-to-end decision support, from data capture to transparent reasoning and communication of AI outputs

@Home:

- Travel, hiking, cooking, games, and DIY 🧑🏻‍🔧





The module team

From left to right:

- Carina (Yu) Zhao (depending on class size)
- Jingda Ying
- Christian Wagner



Lecture Outline

- Module (Course) Overview
- Fuzzy Sets and System—why and how



Part 1:

Module (Course) Overview: COMP4033



Module Aim and Objectives – COMP4033

- **Aim**

- Provide a thorough understanding of fuzzy sets and systems, and associated techniques for handling vague and uncertain data from a theoretical and practical perspective

- **Objectives**

- Introduce the theory and principles of fuzzy sets and fuzzy systems
- Explain how fuzzy and related methods can be used to handle uncertainty in real world examples
- Convey the properties and concepts underlying fuzzy inference systems and their applications
- Explore current research questions, including on explainable AI
- Provide practical experience on the design and implementation of fuzzy systems



Part 2: Module Structure

Level 4 Module



Computing and Labs

- **Sessions:**

- Mondays 900-1000
- Thursdays 1300-1400
- Check timetable AND Moodle page for updates

- **Lab Focus:**

- Student-led exchange
- Practical issues and feedback
- Coursework support





Audience and Expectations

- **Level 4 Module**
 - Final Year BSc and MSci Computer Science students
 - MSc Computer Science students
- **Programming skills are required**
 - **lab sessions will NOT focus on programming support**
- **Interaction and engagement:**
 - We will adapt module content as far as possible based on feedback and interest: the more you put in, the more you get out
 - Lecture notes are 'skeleton resource' – it is expected that you identify and consult other sources, including scientific papers, books, etc. (some additional sources will be way-pointed, e.g. on Moodle)
 - Student collaboration and discussion is expected, EXCEPT for the coursework
- **Sessions will be live/in-person.**



Part 3: Getting Started

Why Fuzzy Sets and Systems



Uncertainty

- Oxford English Dictionary defines the word “uncertainty” as:
 - a) “The quality of being uncertain in respect of duration, continuance, occurrence, etc.; liability to chance or accident. Also, the quality of being indeterminate as to magnitude or value; the amount of variation in a numerical result that is consistent with observation. Something of which the occurrence, result, etc., is uncertain.
An uncertain gain or emolument.
 - b) The state of not being definitely known or perfectly clear; doubtfulness or vagueness.
Law. In phrase: bad, or void, for uncertainty.
Something not definitely known or knowable; a doubtful point.
 - c) The state or character of being uncertain in mind; a state of doubt; want of assurance or confidence; hesitation, irresolution.
In phrase: at uncertainty, upon uncertainties.
 - d) Economics. (The quality of) a business risk which cannot be measured and whose outcome cannot be predicted or insured against (see quotes. 1921 and 1964).
 - e) (Heisenberg's) uncertainty principle (Physics), a principle of quantum mechanics implying that certain pairs of observables (e.g. the momentum and position of a particle, the energy and lifetime of a quantum level) cannot both be precisely and simultaneously known, and that as one of any pair is more exactly defined, the other becomes more uncertain. The principle is usually stated as an inequality such that the product of the uncertainties of the pair of observables cannot be less than a quantity of the order of Planck's constant.”
- Is there a clear definition of ‘uncertainty’?
- Which, if any of these, relate to uncertainty in a probabilistic and, or a ‘fuzzy’ sense?



If you are interested, we can spend more time discussing the nature of uncertainty.

Thought-experiment: how would you program this statement?



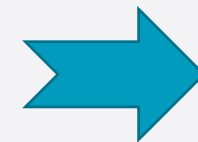
Uncertainty in Science – in (very) brief

- **~Pre-20th century: broadly a view of uncertainty-as-a-flaw in (western, physical) science**
 - *“In physical science a first essential step in the direction of learning any subject is to find principles of numerical reckoning and practicable methods for measuring some quality connected with it. I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind; it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of science, whatever the matter may be.”*
Scottish physicist and mathematician William Thomson (better known as Lord Kelvin): W. T. Kelvin, Popular Lectures and Addresses, Macmillan & Co. 1891
- **19th century: the rise of statistics and probability theory—and recognition of uncertainty as essential and useful**
 - 1871, Austrian physicist Ludwig Boltzmann introduces what becomes known as statistical thermodynamics, a theory developed initially to study the motions of gas molecules in a closed space.
 - This is soon followed by the establishment of a new field of physics which applies the same statistical reasoning in a mechanical context: statistical mechanics.
 - The main difference is the use of statistics (and probability theory) in order to avoid the problems of Newtonian mechanics which were not suitable for the complex systems (at this microscopic level) at hand.



Uncertainty in Science – in (very) brief

- **20th century: evidence for the inevitability of uncertainty**
 - 1926: Werner Heisenberg's Principle of Uncertainty.
 - 1955: "*Scientific knowledge is a body of statements of varying degrees of certainty – some most unsure, some nearly sure, but none absolutely certain.*" Richard P. Feynman
- **General focus on systems with a large number of components and a high degree of randomness**
 - For more detail, see for example: G. J. Klir, "Uncertainty and Information, Foundations of Generalized Information Theory", Wiley, 2006.
- **Hardware-led development of multi-valued logics**
 - 1963: Wilkinson publishes on multi-valued logic based on set theory, implementing for example a 'fuzzy' logic conjunction (OR) and disjunction (AND) through 'analogue logic'.
- **Fuzzy Sets – focus on ambiguity, imprecision**
 - 1965: [Lotfi Zadeh publishes 'Fuzzy Sets' in Information and Control, vol.8, pp. 338-353](#)
 - Discussion:
 - What did you think of the paper?
 - What did you get out of it?
 - Question: is the concept of a fuzzy set statistical in nature?
 - In-class presentations of papers – what are your thoughts?



**Not today... but common
for Level 4 courses.**



Getting into it...

Sets, Boolean Logic, and beyond



Sets—recap

- **Set:** a collection of objects or elements, for example:
 - the characters in the Roman alphabet described as a list:
$$R = \{a, b, c, \dots, y, z\}$$
 - the set of natural numbers (incl. 0): $\mathbb{N} = \{0, 1, 2, \dots\}$
- **Set membership:** an element x either is, or is not, a member of a given set S
 - $x \in S$, or $x \notin S$
- An element cannot be in the same set twice and the order of elements does not matter
 - $S = \{1, 2, 3\} = \{1, 2, 2, 2, 3\} = \{1, 3, 2\}$



Sets—recap

- **Set equality:** two sets are equal if and only if they have the same elements.
- **Basic set operations,** given two sets:
 $A = \{a, b, c, d, e\}$ and $B = \{d, e, f, g\}$, we have:
 - **Union:** $A \cup B = \{a, b, c, d, e, f, g\}$, i.e. the set containing all elements of A and B .
 - **Intersection:** $A \cap B = \{d, e\}$, i.e. the set containing elements that are both in A and in B .
 - **Set Difference:** $A \setminus B = \{a, b, c\}$, i.e. the set containing all elements belonging to A , but not B .
 - **Etc.**



Boolean Logic—recap

- Two truth values: true, false
- Logical Connectives: not, and, or

Name	Symbol	Explanation
Negation	$\neg$	Not
Conjunction	$\wedge$	And
Disjunction	$\vee$	Or
Implication	$\Rightarrow$	If... then...
Equivalence	$\Leftrightarrow$	If and only if...



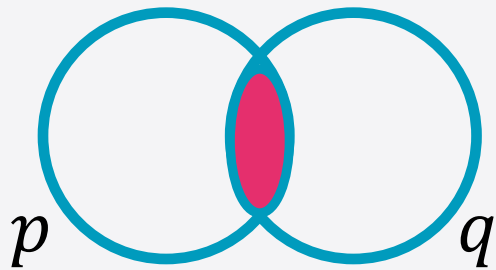
Boolean Logic—recap

- $p \vee q \Leftrightarrow q \vee p$: Commutativity
- $p \wedge q \Leftrightarrow q \wedge p$: Commutativity
- $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$: Associativity
- $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$: Associativity
- $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$: Distributivity
- $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$: Distributivity
- $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$: First De Morgan law
- $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$: Second De Morgan law

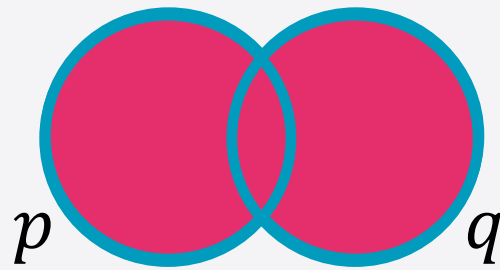


Boolean Logic and Sets

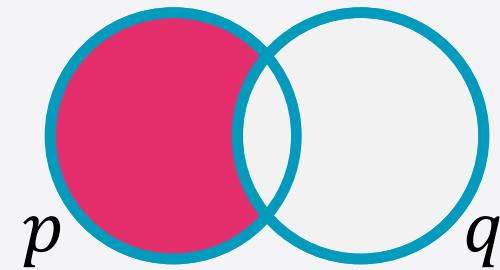
p and q



p or q



p and not q





Characteristic or membership functions

- The membership of an element in a set can be described by a characteristic function – often referred to as a membership function.
- For a given set A , this function assigns a value $\mu_A(x)$ to every x in the universal set X , such that:
 - $\mu_A(x) = 1 \iff x \in A$
 - $\mu_A(x) = 0 \iff x \notin A$
- In other words, the characteristic function maps elements of the universal set to the set $\{0,1\}$:
 - $\mu_A(x): X \rightarrow \{0,1\}$



Sorites paradox

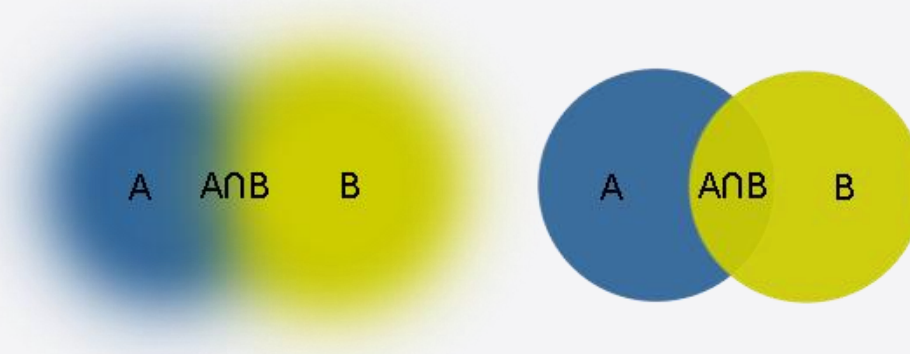
- Sorites is derived from the Greek word for 'heap'
- Consider a heap of sand with, say, 1,000,000 grains of sand
- If you remove a grain of sand, is it still a heap of sand?
- What if you remove another one?
- And another one...
- And...
- Similar: 'bald man paradox' and many others...





Fuzzy Venn diagrams?

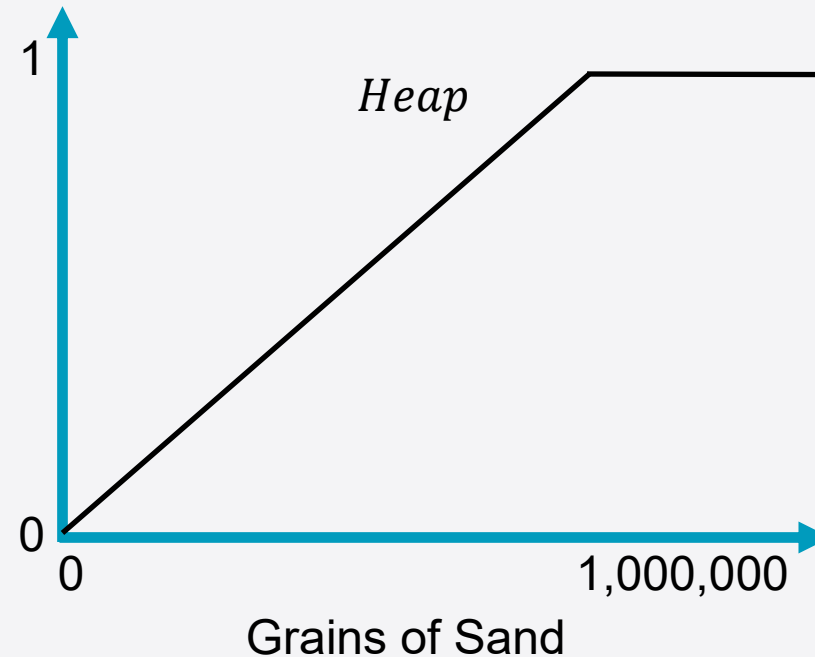
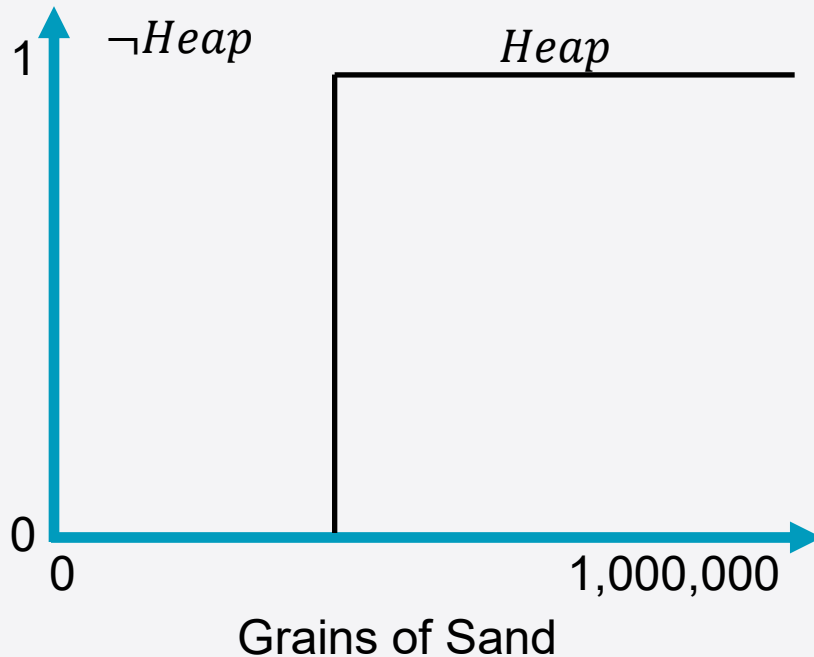
- How do these relate to classical Venn diagrams?
- How does this relate to the 'heap'?





Using our characteristic function to describe heaps...

What would one better description of the heap look like?





We need a different characteristic function!

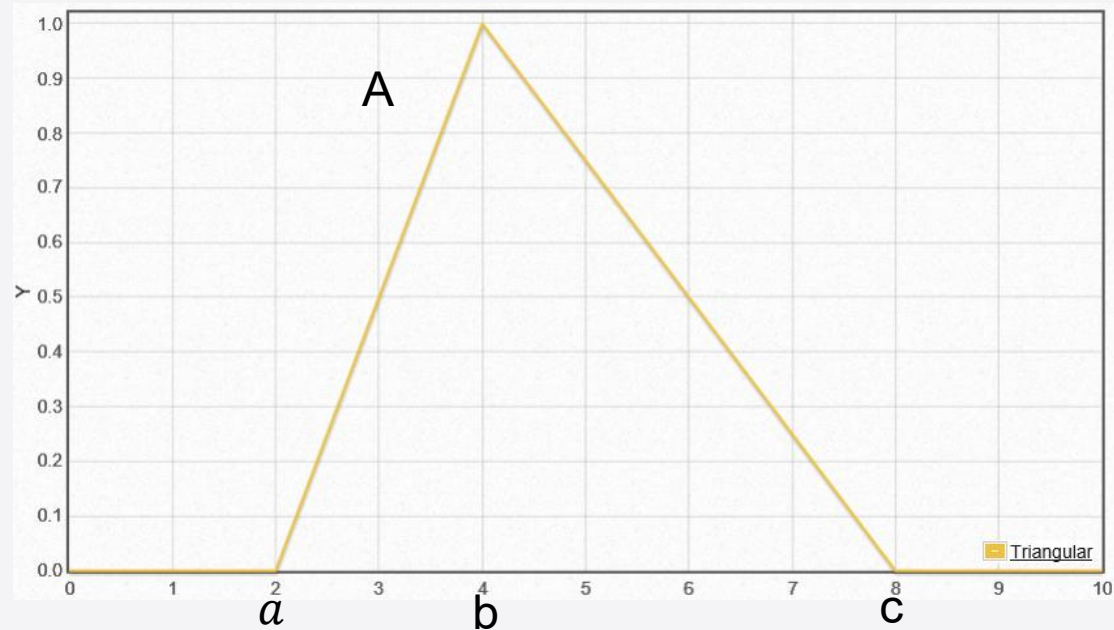
- Beyond mapping to two states, i.e. $\{0,1\}$
- Map elements from the universal set to a real value in $[0,1]$, i.e. for the set *Heap*, we have:
 - $\mu_{Heap}(x): X \rightarrow [0,1]$
- So what is the membership in the set *Heap* of for example:
 - 2 grains of sand
 - 200,000 grains of sand



Example Characteristic Function

- Triangular Membership Function
- Three parameters a, b, c

$$\mu_A(x) = \begin{cases} 0, & x \leq a \\ 0, & x \geq c \\ \frac{x-a}{b-a}, & x \in [a, b] \\ \frac{c-x}{c-b}, & x \in [b, c] \end{cases}$$

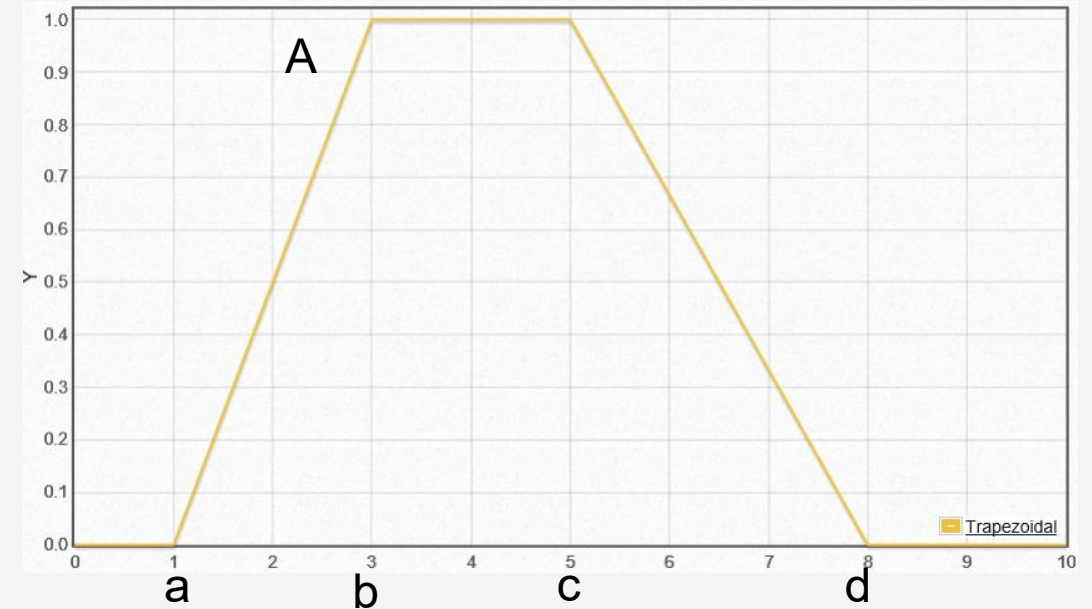




Example Characteristic Function

- Trapezoidal Membership Function
- Four parameters a, b, c, d

$$\mu_A(x) = \begin{cases} \frac{x-a}{b-a}, x \in [a, b] \\ 1, x \in [b, c] \\ \frac{d-x}{d-c}, x \in [c, d] \\ 0, \text{otherwise} \end{cases}$$



- Of course, there are many other characteristic or membership functions, including Gaussian, Bell, etc.
- Membership functions describe fuzzy sets, and we often refer to them by the same name for convenience, for example: 'trapezoidal fuzzy set'



Quick 'break'...

Kahoot!



We have fuzzy sets...

Example application – rule-based inference



If...Then Rules in Fuzzy Systems

- Inference is performed by utilising a set of rules connecting premises to conclusions.
 - premise (the 'if ...' part) is called the antecedent(s)
 - conclusion (the 'then ...' part) is called the consequent(s)
- Inference for fuzzy sets is simplified by putting aside several formal considerations of logical implication, for example:
 - Law of excluded middle, i.e. either a proposition is true, or its negation is true (see [Law of excluded middle - Wikipedia](#)), more formally:
 - $A \cup \bar{A} = X$
 - Law of contradiction, a proposition and its negation cannot both be true, formally:
 - $A \cap \bar{A} = \emptyset$
 - Note: fuzzy sets enable the consideration of edge cases, e.g. a 'red apple'



A toy example

- **IF** it is *cold* **THEN** wear *many layers of clothing*



Why and where is this useful?

Early applications in Control...



Fuzzy Inference – Applications

- First applications in control appeared quickly:
 - 1975: Cement Kiln in Denmark
 - 1985: Sendai Railway in Japan (braking, acceleration)
- Powerful and efficient means to dealing with sensor uncertainty while leveraging human insight
- Traditionally, Fuzzy Logic based applications have been embraced in Asia – less so in the West.
- However, today there are a vast number of applications in control and beyond.



VW DSG gearbox



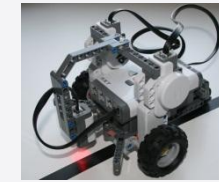
Panasonic FL rice cooker



Samsung FL washing machine



CNC milling



...and of course - robots



Fuzzy Inference – Applications

- And of course, applications beyond control, such as in respect to entity linking and knowledge graphs...

Palantir

Get Started

Overview

Entity Resolution (ER) is a critical data management process that involves identifying and linking records across different data sources that refer to the same real-world entity, such as a person, organization, or product.

This process is essential for ensuring data consistency and accuracy, especially in environments where data is collected from multiple, disparate sources.

By resolving entities accurately, organizations can eliminate duplicate records, reduce errors, and enhance data-driven insights.

Palantir Entity Resolution

Palantir Entity Resolution provides continuous matching of millions of records from disconnected systems with end-to-end visibility into each step of the resolution process, enabling your organization to deploy an entity resolution and graphing solution to gain a de-duplicated data foundation of high trust. Palantir Foundry overcomes the challenges associated with entity resolution for massive datasets by using hashing methods and AI/ML models to improve speed with reduced computational spend. Fuzzy matching techniques further improve accuracy.



Read CEO Alex Karp's Letter to Shareholders

Palantir

Get Started

Palantir Entity Resolution

Seamlessly link records with AI to establish a reliable, de-duplicated data foundation.

ERP Migration

[ERP Migration] Entity Resolution Match Labelling

Individual Match Review | Grouped Match Review | Bulk Review

Match Selection

Filter Matches

Matches (8,644)

Match Score

Electronic Company-Electronic Company

Match Priority: High

Match Score: 100%

Review Status: Confirmed

ER Score Table: Supplier Deduplication

H.A.G. Potsdam Gr.17-H.A.G. Potsdam Gr.18

Match Priority: High

Match Score: 100%

Review Status: Confirmed

ER Score Table: Supplier Deduplication

Spedition Gruber-Spedition Gruber

Match Priority: High

Match Score: 100%

Review Status: Confirmed

ER Score Table: Supplier Deduplication

Abbot Supplies Inc.-Abbot Supplies Inc.

Match Priority: High

Match Score: 100%

Review Status: Confirmed

ER Score Table: Supplier Deduplication

Sapsota Company Limited-Sapsota Company Limited

Match Priority: High

Match Score: 100%

Review Status: Confirmed

ER Score Table: Supplier Deduplication

Review Match

Unreviewed Matches: 5592 | Confirmed Matches: 3026 | Rejected Matches: 26

100% Overall Score

Confirmed

Predicted Label Confirmation Probability: 99%

Match Priority: High

Review Status: Confirmed

Match At: Mon, Nov 13, 2023, 9:07 PM

ER Score Table: Supplier Deduplication

Left Entity

Source Table: Vendor Master (Lst)

H.A.G. Potsdam Gr.17

ERP Migration | Supplier

Overview | Properties

Properties

City ont01	Potsdam
Country land1	DE
Name Of Representative J 1kfrepe	No value
Name name1	H.A.G. Potsdam Gr.17
Postal Code Pstlz	14467
Street Straz	Schopenhauerstraße 38
Telephone 1 tel1	0331/918-0
Vat Registration No Stecg	No value
Vendor Lifnr	T-K500C17

Right Entity

Source Table: Vendor Master (Lst)

H.A.G. Potsdam Gr.18

ERP Migration | Supplier

Overview | Properties

Properties

City ont01	Potsdam
Country land1	DE
Name Of Representative J 1kfrepe	No value
Name name1	H.A.G. Potsdam Gr.18
Postal Code Pstlz	14467
Street Straz	Schopenhauerstraße 38
Telephone 1 tel1	0331/918-0
Vat Registration No Stecg	No value
Vendor Lifnr	T-K500C18



Mamdani Inference

One of the most famous uses of fuzzy sets...



Mamdani inference

- Mamdani introduced the first successful form of fuzzy inferencing in 1975
 - E.H. Mamdani and S. Assilian, “*An experiment in linguistic synthesis with a fuzzy logic controller*,” International Journal of Man-Machine Studies, vol. 7, no. 1, pp. 1–13, 1975.
- A controller for a steam engine



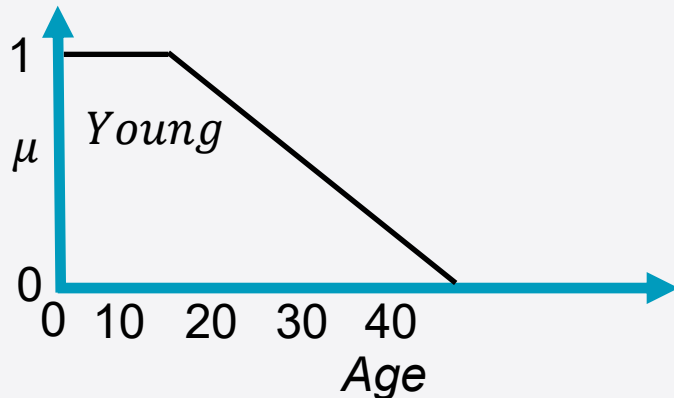
Methodology

- Describe a system as a set of rules:
 - *IF x_1 is A_1 (AND/OR) x_2 is B_1 ... THEN y is C_1*
- **For each rule:**
 - for each antecedent, evaluate the membership degree of the crisp input value in the fuzzy term (e.g. $\mu_{A_1}(x_1)$);
 - combine all membership degrees using appropriate fuzzy operator (using a t-norm, such as minimum or product; or t-conorm, such as maximum);
 - fire the rule at the strength (combined degree) to obtain the rule output based on the rule's consequent.
- **Aggregate the output of each rule** and interpret in some way (e.g. defuzzify the fuzzy output to a crisp value).



Worked Example – step-by-step

- Rule 1: IF Tall AND Young THEN basketball_careerpotential is good
- Rule 2: IF Short THEN basketball_careerpotential is poor
- Rule 3: IF Tall AND Middle-Aged THEN basketball_careerpotential is ok
- Consider a trapezoidal membership function for young:



Note: a common notation for degrees of membership is $\mu(x)/x$
This does NOT mean division.
We can use it to describe a membership functions in discrete terms, using +.
This does NOT mean addition.
For example:
 $\text{young} = 1/0 + 1/10 + .75/20 + .5/30 + .25/40$



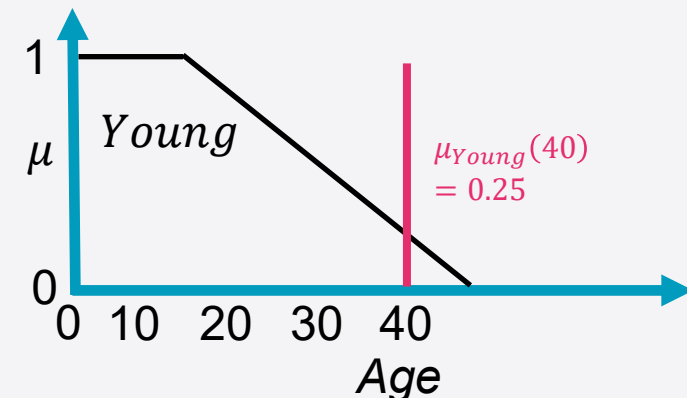
One Antecedent...

Step by step



Determining the degree of truth of an antecedent

- Given a numeric input x' , for example, age $x' = 40$.
- First, we convert the numeric input into a fuzzy set using a process called fuzzification
- Fuzzification: turn a numeric input x' into a fuzzy set X'
 - Here, we use singleton fuzzification, i.e. each number x' becomes a 'spike', i.e. a set where:
 - $\mu_{X'}(x) = \begin{cases} 1, & \text{if } x = x' \\ 0, & \text{otherwise} \end{cases}$
- To determine the degree of truth given an antecedent, we compute the intersection and use its (maximum) degree of membership at x'



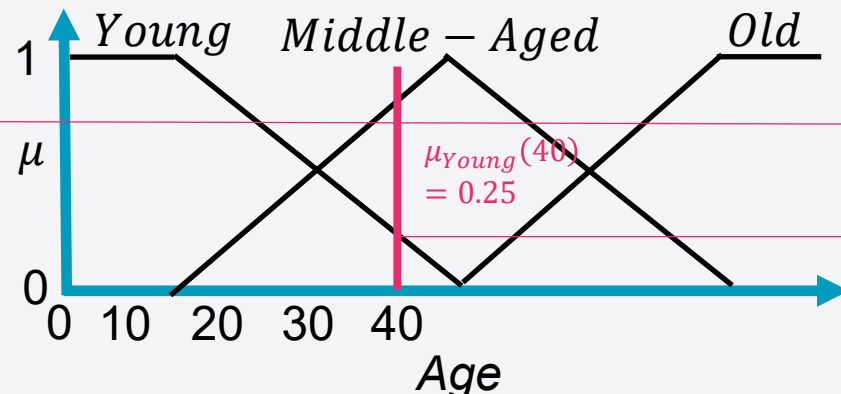
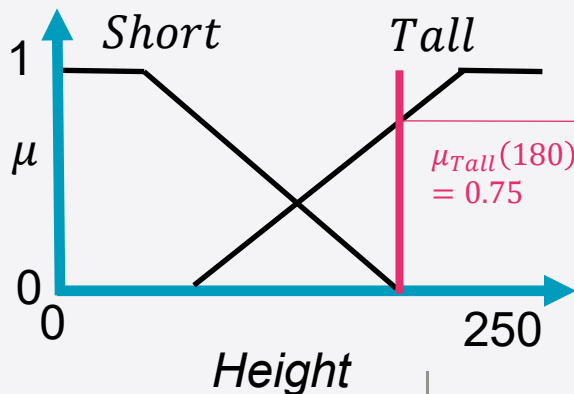


Multiple antecedents...



Example: Firing Degree/Strength of a rule

- Determining the degree of truth of the antecedent of a rule, also known as it's the rule's *firing degree*.
- Rule 1: IF Tall AND Young THEN basketball_careerpotential is good
 - Example: Height = 1.8m, Age = 40
 - Establish degrees of membership of the input (sets) for each antecedent set
 - The rule uses conjunction, thus use a t-norm $\star$ (e.g. min) to combine the degrees of membership within the antecedent



Firing Strength of Rule 1:

$$\begin{aligned}\mu_{Tall}(180) \star \mu_{Young}(40) \\ &= 0.75 \star 0.25 \\ &= \min(0.75, 0.25) \\ &= \mathbf{0.25}\end{aligned}$$

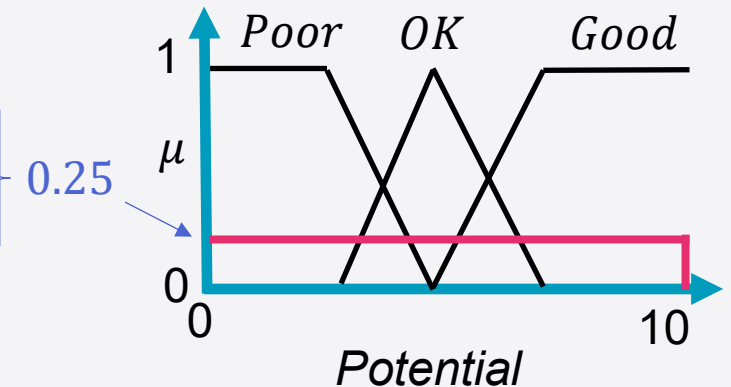
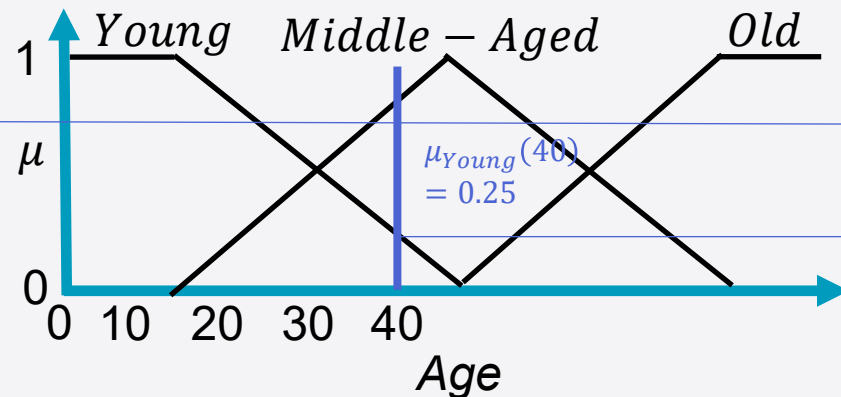
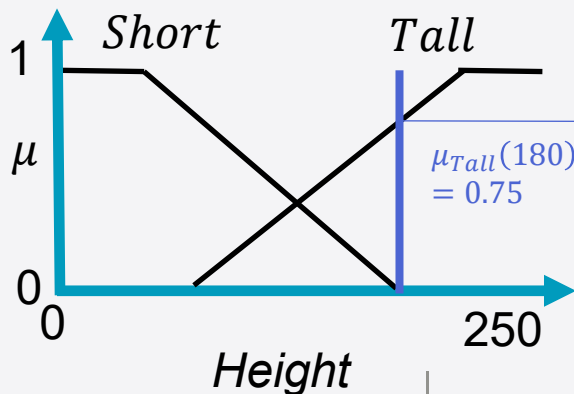


A rule with a consequent...



Example: Firing a consequent of a rule

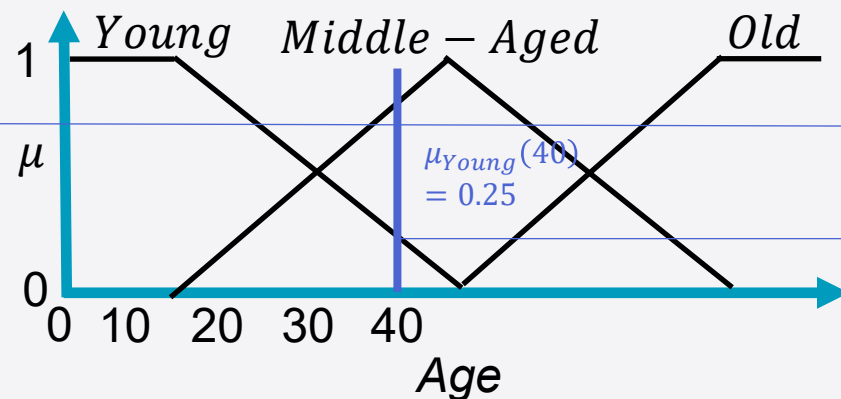
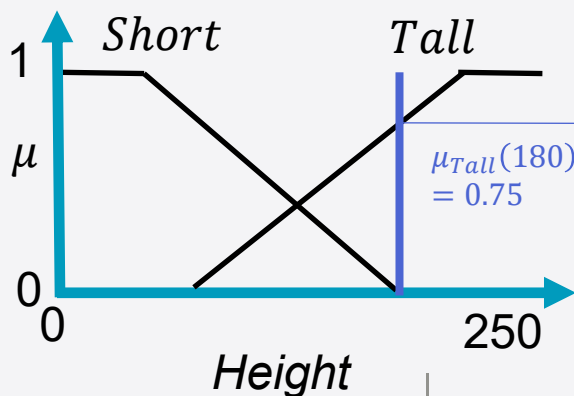
- Use the firing strength of the antecedent to determine the degree of truth of the consequent.
- Rule 1: IF Tall AND Young THEN basketball_careerpotential is good
 - Our example: firing strength: 0.25
 - Model firing strength as a **cylindrical extension C**
 - Compute the output set as the intersection of C and the consequent *Good*



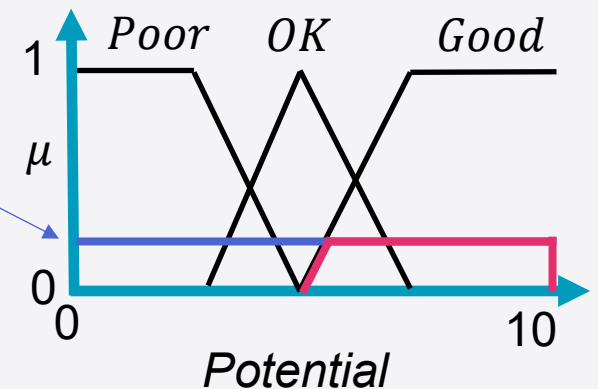


Example: Firing a consequent of a rule

- Consider:
 - $Good: 0/5 + 0.1/6 + 0.5/7 + 1/8 + 1/9 + 1/10$
 - Firing Str. Cyl. Ext. $C_1: 0.25/5 + 0.25/6 + 0.25/7 + 0.25/8 + 0.25/9 + 0.25/10$
 - Rule 1: $C_1 \cap Good = 0/5 + 0.1/6 + 0.25/7 + 0.25/8 + 0.25/9 + 0.25/10$
 - Note: here we use the t-norm *minimum*



0.25





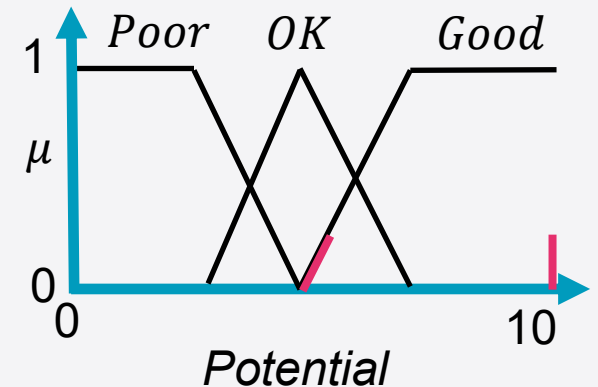
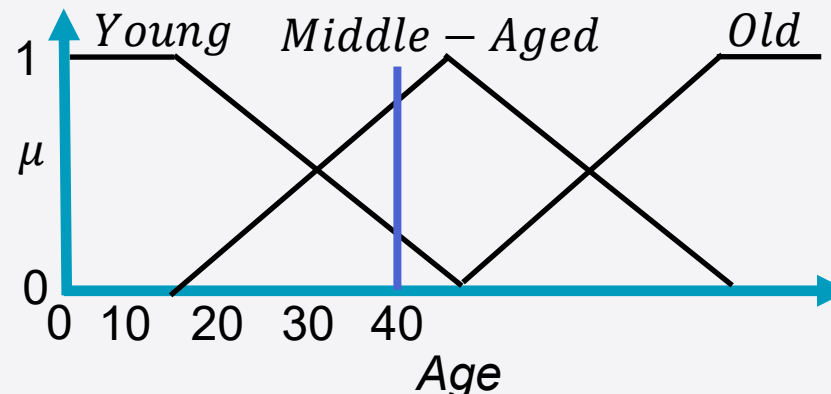
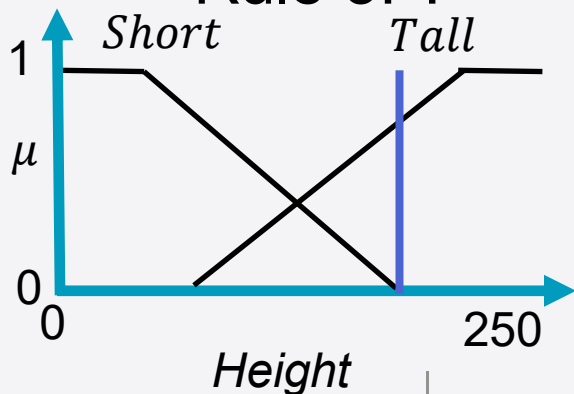
Question

How about multiple rules?



Example: Multiple rules

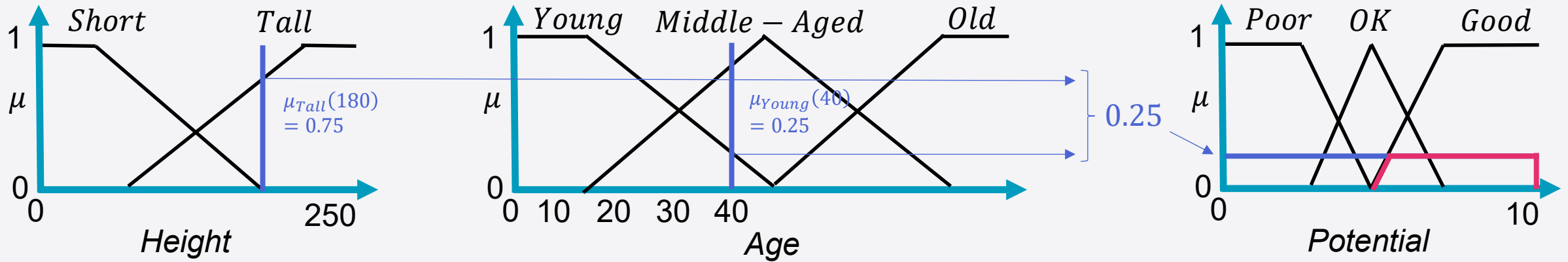
- Our rule base:
 - Rule 1: IF Tall AND Young THEN basketball_careerpotential is good
 - Rule 2: IF Short THEN basketball_careerpotential is poor
 - Rule 3: IF Tall AND Middle-Aged THEN basketball_careerpotential is ok
- Example: Height = 1.8m, Age = 40
 - Rule 1: done
 - Rule 2: ?
 - Rule 3: ?



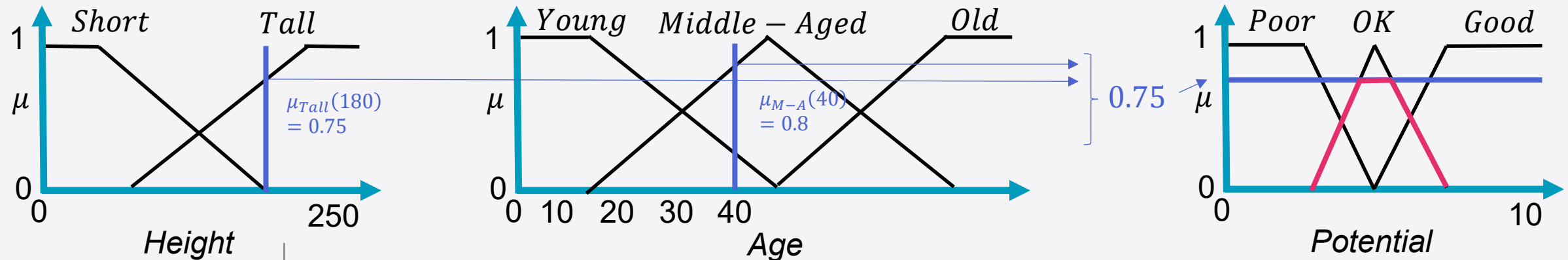


Example: Rule 3

- Rule 1: IF Tall AND Young THEN basketball_careerpotential is good



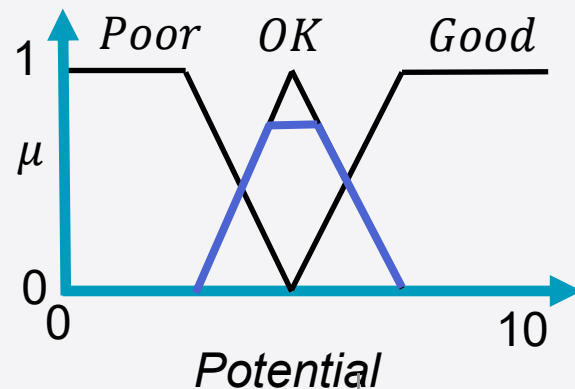
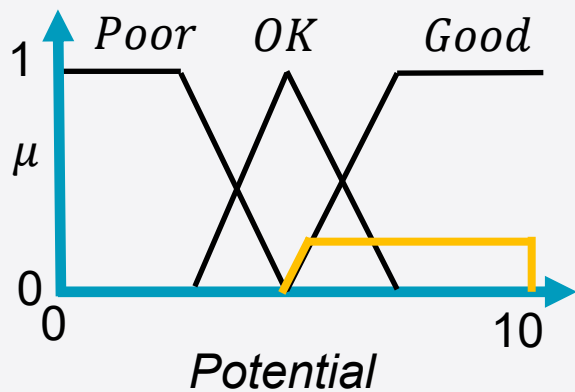
- Rule 3: IF Tall AND Middle-Aged THEN basketball_careerpotential is ok





Example: Combine rule outputs

- Outputs from individual rules are combined using disjunction, i.e. the union of both rules' outputs, operationalized using a t-conorm, such as *maximum*.



- Outputs from individual rules are combined using a t-conorm, such as *maximum*.
 - $R_1 = C_1 \cap \text{Good} = 0/5 + 0.1/6 + 0.25/7 + 0.25/8 + 0.25/9 + 0.25/10$
 - $R_2 = : \emptyset$
 - $R_3 = C_2 \cap \text{Ok} = 0/3 + 0.7/4 + 1/5 + 0.7/6 + 0.0/7 + 0.0/8$
- $R_1 \cup R_2 \cup R_3 = \max(\mu_{R_1}(y), \mu_{R_2}(y), \mu_{R_3}(y)) = 0/3 + 0.7/4 + 1/5 + 0.6/6 + 0.25/7 + 0.25/8 + 0.25/9 + 0.25/10$



Defuzzification

So now what?



Defuzzification



In general, the result of Mamdani inference is a complex output. It is a distribution, and more specifically, the membership function characterising the output fuzzy set for the given inputs.



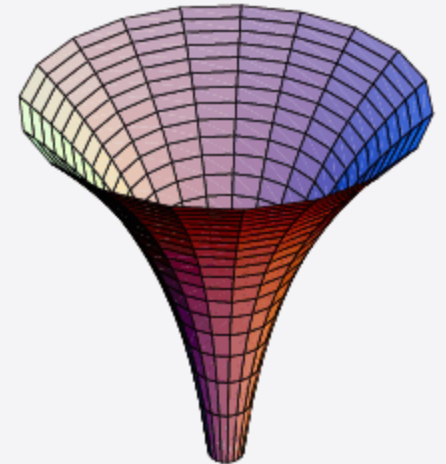
The process of converting a fuzzy set to a number is called 'defuzzification'.



Often, for example in control applications (think: steering a quadcopter), a single (crisp) output is required, i.e. a number.



Mamdani chose to use a method whereby the centre of the area under the output set is used called the centroid, reflecting the centre-of-gravity of the distribution.

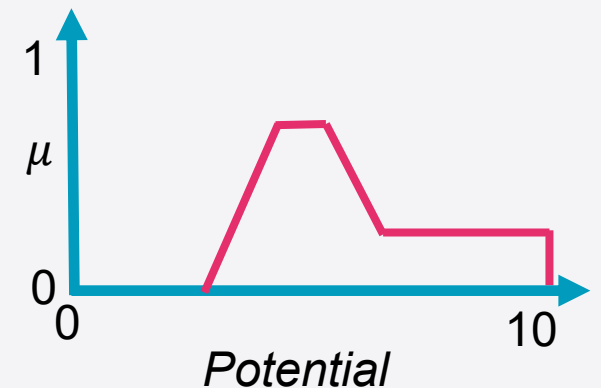


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Defuzzification – the general problem

- The result of Mamdani inference is an ‘arbitrary’ output fuzzy set(s)
- What is the meaning of such a set?
- Two principal approaches:
 - Numeric defuzzification—a mentioned, for example the centroid, or the mean-of-maxima, or height.
 - Linguistic defuzzification: a linguistic term representing the output set if found, using a similarity and/or distance metric.





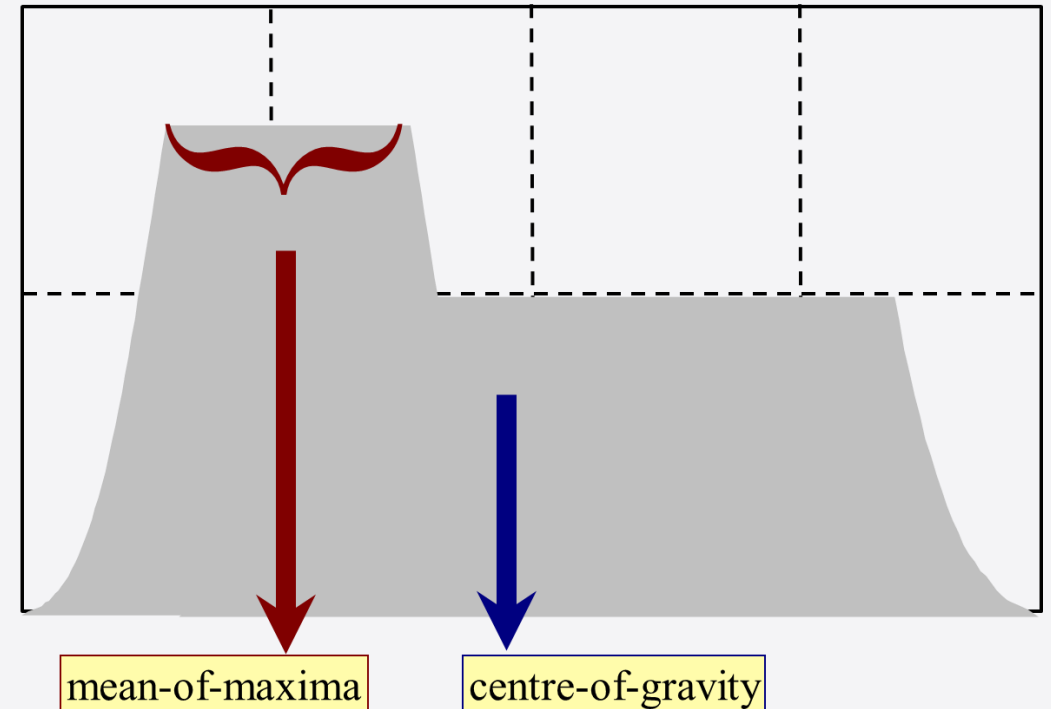
Numeric Defuzzification:

- Mean of Maxima (MoM)
 - $x_{MoM} = \text{mean}(\text{argmax}_x \mu(x))$
 - In our example: $x_{MoM} = 5$
- Centroid (Cen) i.e. centre of gravity

- $x_{Cen} = \frac{\sum_{i=1}^N x_i \mu(x_i)}{\sum_{i=1}^N \mu(x_i)}$

- In our example:

$$x_{Cen} = \frac{3 * 0 + 4 * 0.7 + 5 * 1 + 6 * 0.6 + 7 * 0.25 + 8 * 0.25 + 9 * 0.25 + 10 * 0.25}{0 + 0.7 + 1 + 0.6 + .25 + 0.25 + 0.25 + 0.25}$$
$$= \frac{19.9}{3.3} = 6.\overline{03}$$



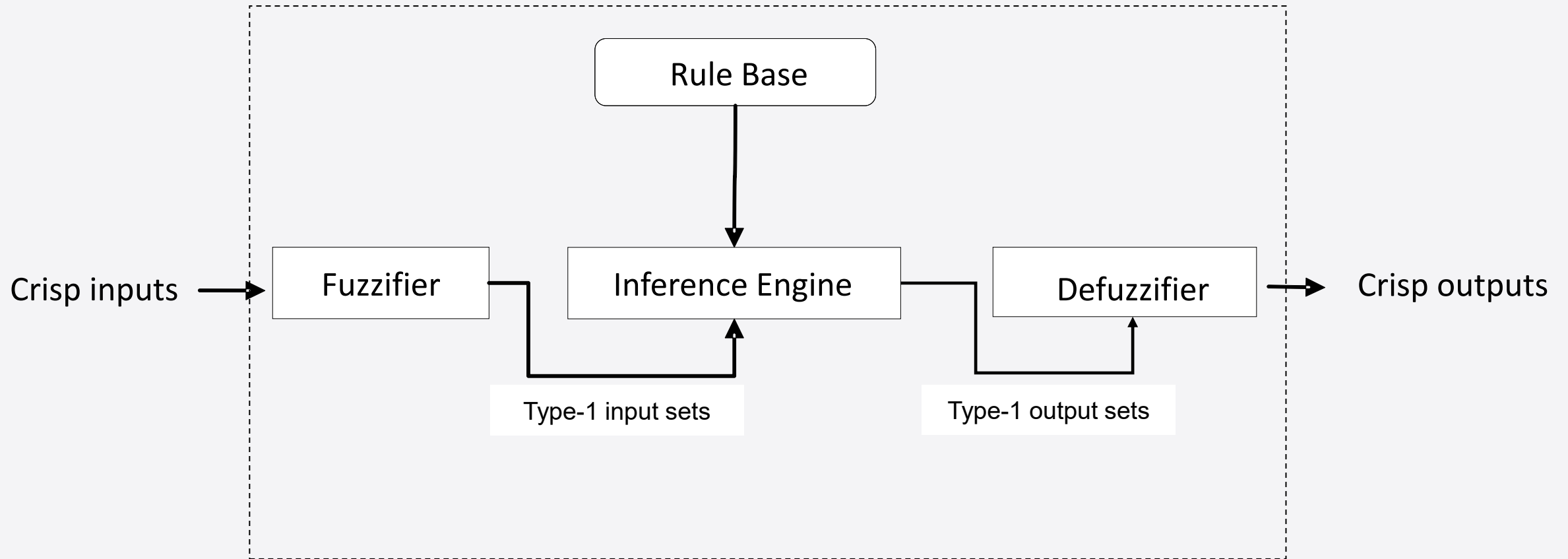


Zooming out

Mamdani Fuzzy Systems



Structure of Mamdani Fuzzy Systems





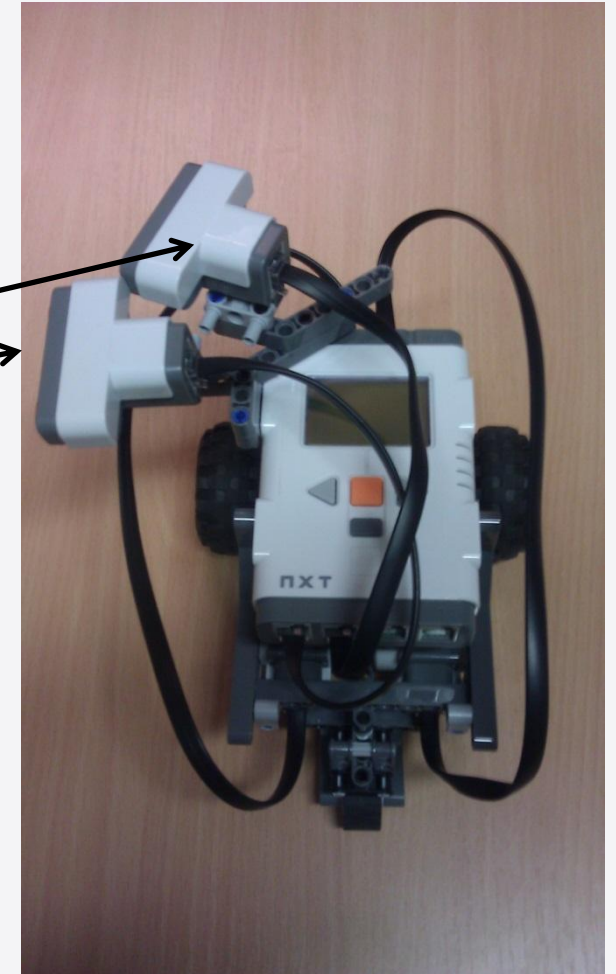
A control example

Robot Navigation



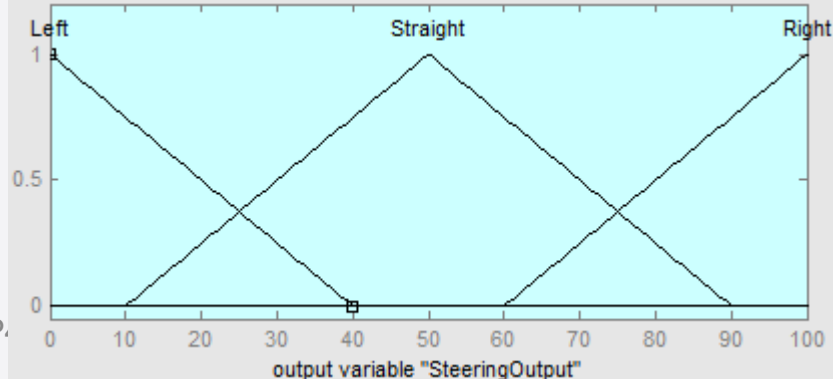
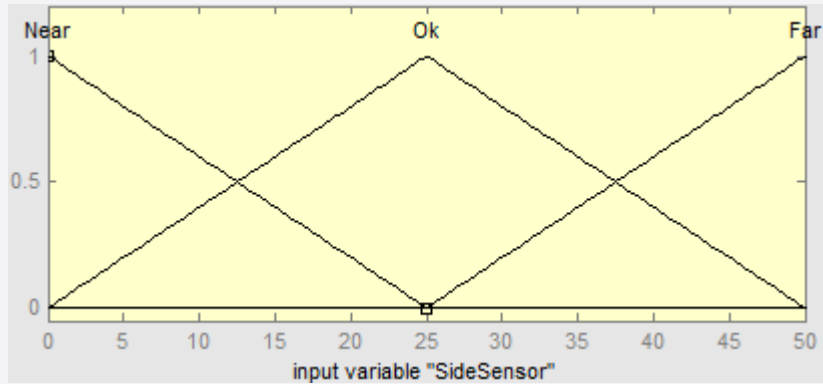
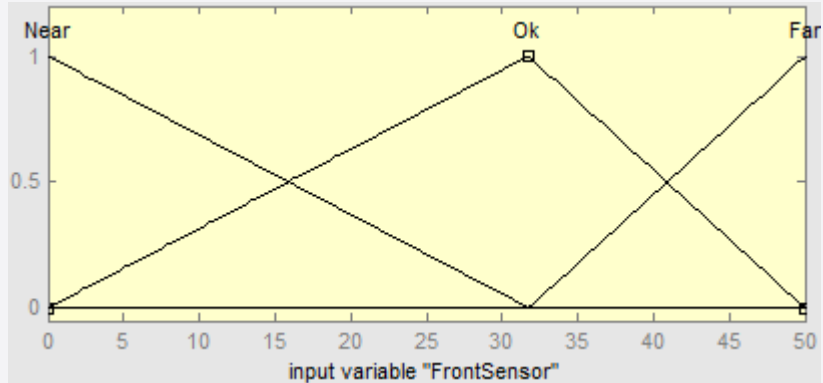
Problem: Left-hand wall-following

- A common control feature of indoor robots
- Desired distance to wall:
 - 25cm
- 2 inputs:
 - front sonar sensor facing the wall at an angle
 - side sonar sensor directly facing the wall
- 1 output:
 - Steering from 0-100 where 0 is hard left and 100 is hard right
- Note, design your own FLCs:
 - Matlab Fuzzy Logic Toolbox (run Matlab and type “fuzzy”)
 - JuzzyOnline: <http://juzzyonline.wagnerweb.net>





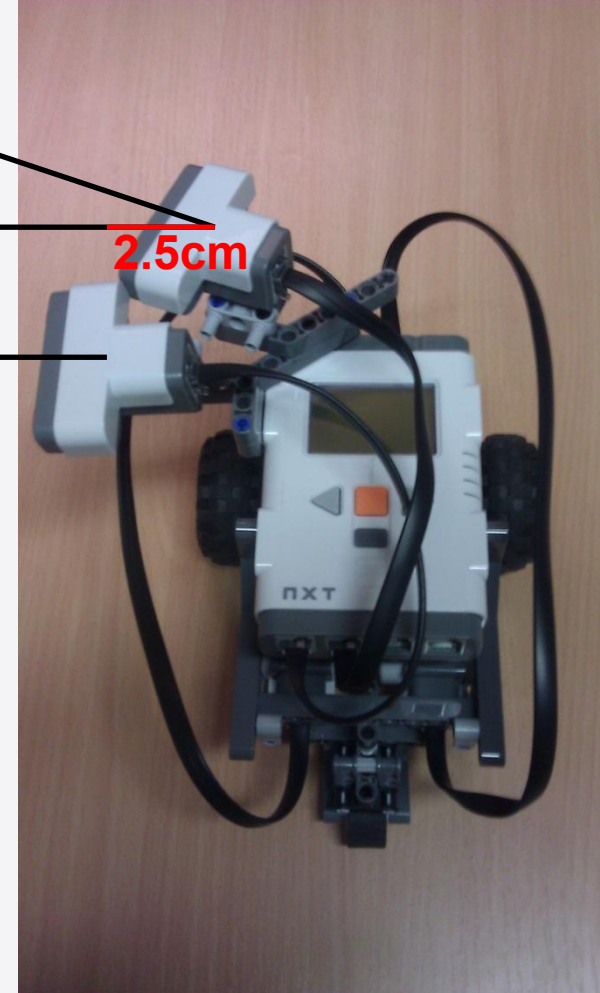
Fuzzy Set Design



Wall

$$d_f = \frac{25 + 2.5}{\cos(30^\circ)} \cong 31.75cm$$

$$d_s = 25cm$$





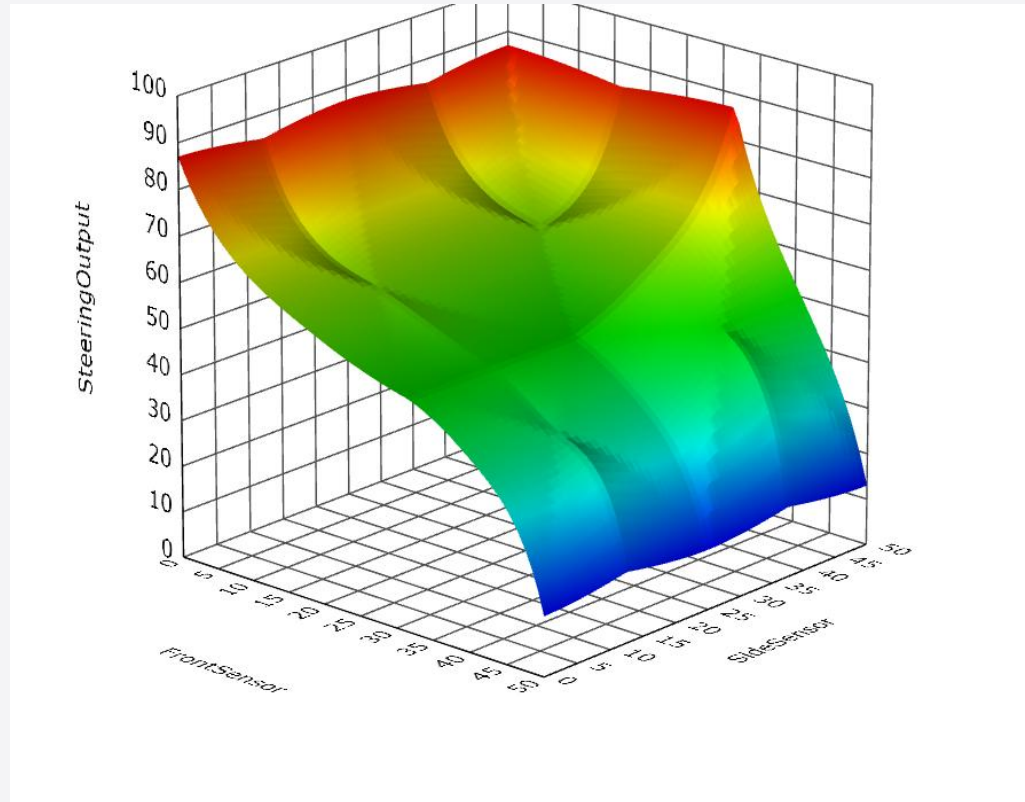
Rule Base Design

Rule Number	Front Sensor	Side Sensor	Steering Output
1	Near	Near	
2	Near	Ok	
3	Near	Far	
4	Ok	Near	
5	Ok	Ok	
6	Ok	Far	
7	Far	Near	
8	Far	Ok	
9	Far	Far	



An output summary

- Called a control surface





Notes and Summary

- You can try designing your own system using for example [JuzzyOnline](#)
 - You can build your own in Matlab, other library or your own code
 - In the COMP4033 module, students pick between R, Java or Python.
- Take-aways on fuzzy sets and systems
 - Fuzzy sets are designed to capture vague and uncertain information, mimicking the type of uncertainty which people use daily (“its quite warm”).
 - Rule-based fuzzy systems are ONE TYPE of application of fuzzy sets—attracting attention (again) at the moment from an explainable AI p.o.v.
 - These systems are knowledge representations, i.e. they are not in of themselves ‘machine learning systems’. However, optimisation and ML can be used to construct and tune fuzzy sets and systems.
 - Many other applications, from clustering, to linguistic summaries, preference elicitation, etc.



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End

Any Questions?